GEGENBAUER PROBABILITY-MEASURE HYPERCONTRACTIVITY

ØISTEIN SØVIK

Evidence is labelled Sketch: this proof has not been posted on arXiv or peer reviewed and should be checked by the reader. AI assistance was used in adapting and checking the argument and exposition.

> Conjecture. The same is true for the measures $c_\alpha(1-x^2)^\alpha\,\mathrm{d}x$ on $(-1,1)$, $\alpha>-1$ (Gegenbauer polynomials).

The preceding sphere conjecture asserts $\|T_\omega\|_{p,q}=1$ for real $-\sqrt{(p-1)/(q-1)}\le\omega\le\sqrt{(p-1)/(q-1)}$, with $1<p\le q<\infty$. Here T_ω multiplies the degree-k orthogonal-polynomial component by ω^k. Write λ for the source's α, and normalize $\mathrm{d}\mu_\lambda(x)=c_\lambda(1-x^2)^\lambda\,\mathrm{d}x$ to have mass one. Source: Janson, "On hypercontractivity for multipliers on orthogonal polynomials," doi:10.1007/BF02384302, §6, conjecture, printed p. 108 (PDF p. 12).

Answered: no. Evidence: Sketch.

No. At $\lambda=-199/200$, $p=2$, $q=4$, $\omega=1/\sqrt{3}$, and $f(x)=x^2-100/101$, the fourth power of the norm ratio is $3983735/3160809>1$.

Proof. Put $d=2\lambda+3>1$. Substituting $y=x^2$ in the beta integral gives

$$\mathbb{E}x^{2k}=\frac{B(k+1/2,\lambda+1)}{B(1/2,\lambda+1)}=\prod_{j=0}^{k-1}\frac{2j+1}{d+2j}.$$

Thus $Q=x^2-1/d$ is orthogonal to constants and to x, so it lies entirely in the degree-two component; hence $T_{1/\sqrt{3}}Q=Q/3$. The moments through degree eight give

$$\mathbb{E}Q^2=\frac{2(d-1)}{d^2(d+2)},\qquad \mathbb{E}Q^4=\frac{12(d-1)(5d^2-13d+12)}{d^4(d+2)(d+4)(d+6)}.$$

Consequently the fourth power of the norm ratio is

$$R(d)=\frac{(d+2)(5d^2-13d+12)}{27(d-1)(d+4)(d+6)},\quad R(d)-1=\frac{-2(11d^3+123d^2+196d-336)}{27(d-1)(d+4)(d+6)}.$$

For $d=101/100$ the cubic is negative, and direct substitution gives the stated rational ratio. All integrals converge because $\lambda>-1$. □

Date: 13 October 2026.